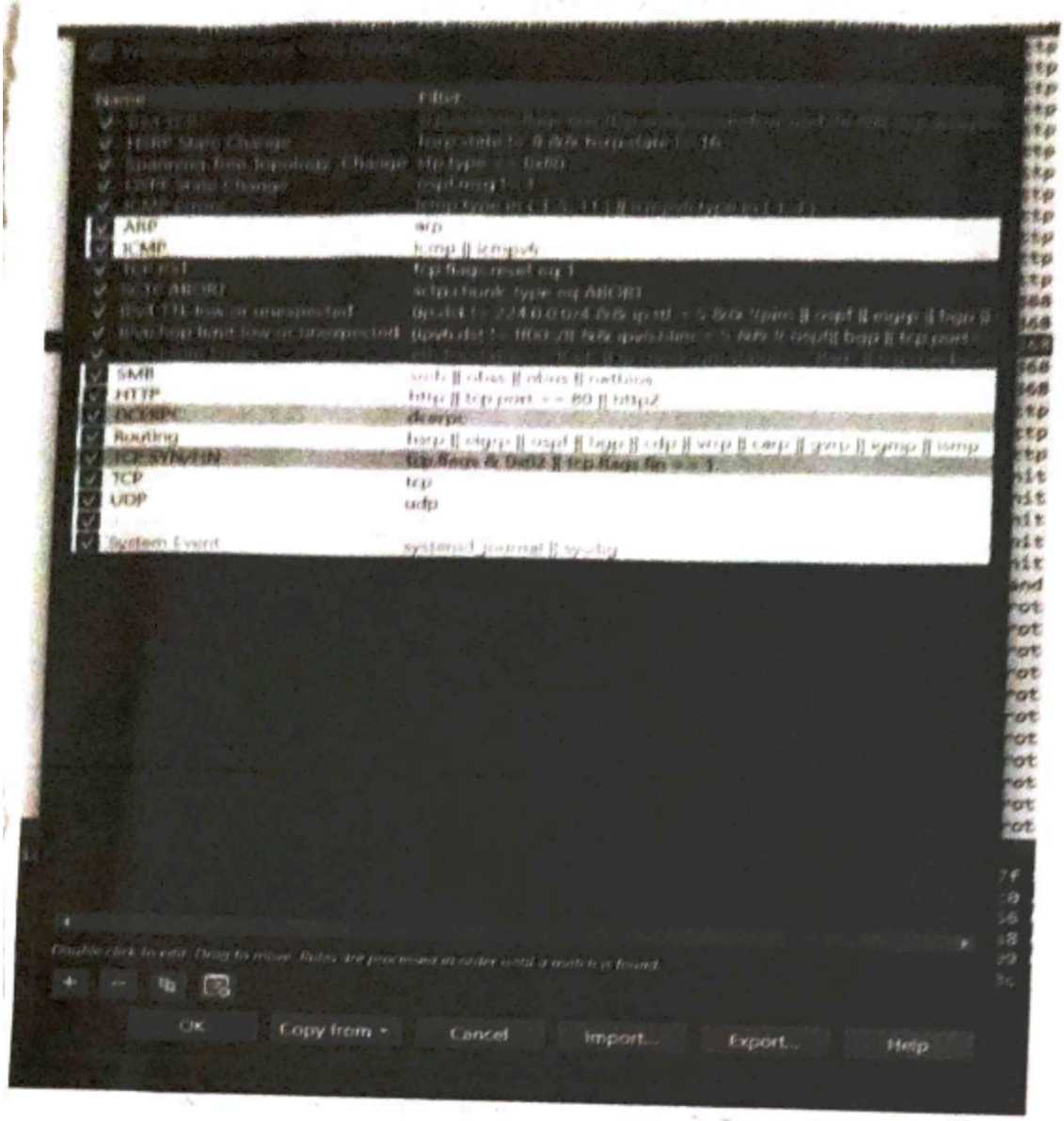
Experiment - 5
Wireshark

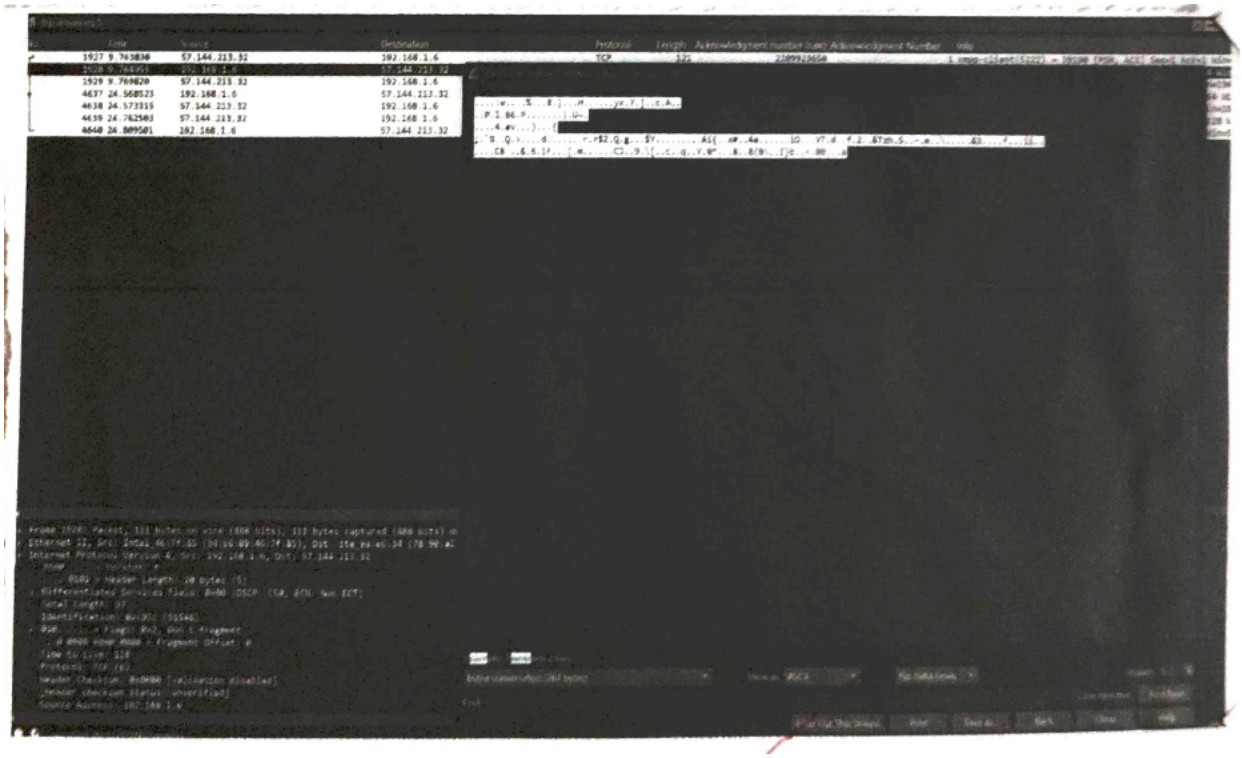
AIM:

To Study the Wireshark packet capture tool and analyse packets at different layers of the protocol stack.

PROCEDURE:

1. Open Wireshark.
2. Select the Local Area Connection / Ethernet interface.
3. Start Packet capture.
4. Capture approximately 100 packets.
5. Stop the capture.
6. Apply filters for:
   - → TCP / UDP
   - → ARP
   - → DNS
   - → HTTP
   - → IP / ICMP
   - → DHCP
7. Select individual packets.
8. Observe the Packet list, Packet Details and Packet Bytes panes.
9. Analyse the Protocol fields.
10. Save the captured packets.


Name
Filter
ARP
arp
ICMP
icmp || icmpv6
SMB
HTTP
http || tcp.port == 80 || http2
DCERPC
dcerpc
Routing
TCP SYN/FIN
TCP
tcp
UDP
udp
System Event
systemd_journal || syslog
OK
Copy from
Cancel
Import...
Export...
Help



The manual specifically gives procedures for filtering TCP/UDP, ARR, DNS, HTTP, ICMP/IP and DHCP packets.

Sample Output:

No.	Protocol	Source	Destination
1.	ARP	192.168.1.10	Broadcast
2	DNS	192.168.1.10	8.8.8.8
3	TCP	192.168.1.10	142.xx.xx.xx
4	HTTP	192.168.1.10	Server.

9/10

Result:

Thus, Wireshark was successfully used to capture, filter and analyze network packets and their protocol fields.